

Introduction to Experimental Design

Foundations of Experimental Thinking

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Learning Objectives

By the end of this lecture, you should be able to:

- Explain why experimental design is critical for valid inference in agriculture.
- Describe the principles of experimentation.
- Distinguish observational studies from designed experiments.
- Spot common design mistakes in agricultural research.

Timing: ~3 minutes

Engagement tips: - Ask for quick examples from their disciplines (crops, livestock, food quality). - Clarify that we focus on concepts today; heavy R starts Wednesday.

When Bad Design Hurts

Caution

Story: A food company tested a new preservative by adding it only to jars processed in the morning shift (cooler temps) and compared shelf life to jars from the hotter afternoon shift. They declared the preservative worked—until it failed in production.

- Confounding: time-of-day and temperature, not the preservative, drove the shelf-life difference.
- Cost: recalls, brand damage, wasted production.
- Lesson: without design, we mistake nuisance variation for treatment effects.

Timing: ~4 minutes

Prompt: “Where do you see similar risks in your work—field gradients, pen location, batch effects?”

Why Agriculture Needs Good Design

- **Horticulture:** Greenhouse bench effects can minimize fertilizer differences.
- **Animal Science:** Pen location alters airflow/heat; confounds diet trials.
- **Crop Science:** Slope, soil moisture, and shade gradients bias yield.
- **Food Science:** Batch-to-batch process drift masks formulation effects.

Timing: ~4 minutes

Metaphor: Randomization = scattering seeds across the field so no treatment hogs the best soil.

Foundations

What Is an Experiment?

- **Plain English:** You **intervene** (apply treatments) and **control** how they’re assigned.
- **Formal:** Assign treatments to experimental units using a rule (often random), then measure responses.
- **Goal:** Isolate treatment effects by managing other sources of variation.

Timing: ~3 minutes

Check-for-understanding: Ask for an example of an experimental unit in their field (plot, pen, batch, plant, animal).

Observational vs. Experimental

- **Observational:** Measure what’s already happening (e.g., soil C vs. yield on farmer fields without previous assignment of treatments). Limited ability to infer causation.
- **Experimental:** You assign fertilizer rates to plots. Randomization lets you claim causation if assumptions hold.
- **Risk:** Treating observational patterns as causal → poor recommendations.

Timing: ~3 minutes

Question: “When might you be forced to use observational data? How does that change your claims?”

Principles of Experimentation



Figure 1: Sir Ronald A. Fisher (1890–1962)

Father of modern DOE

Fisher's Three Pillars:

- **Replication:** Multiple units per treatment to estimate variability and improve precision
- **Randomization:** Fair assignment eliminates bias and ensures valid statistical inference
- **Blocking:** Group similar units to control known variation before comparing treatments

Fisher also pioneered **factorial designs** and developed **ANOVA** for analyzing experimental data.

Timing: ~6 minutes

Key points: - Emphasize Fisher's foundational contributions to experimental design - These three principles are still the cornerstone of DOE today Fisher introduced these concepts in the 1920s-1930s while working at Rothamsted Agricultural Station in England.

Historical context: Fisher was in his early 30s (age 32–40) when he developed the three principles of experimentation at Rothamsted (major publications: 1926–1935).

Analogy: - Randomization: like scattering seed evenly instead of planting the best seed only in the best corner. - Blocking: like grouping plots by slope before randomizing within each band. - Replication: having multiple independent observations to separate signal from noise.

Consequences of Poor Design

Confounding in Action

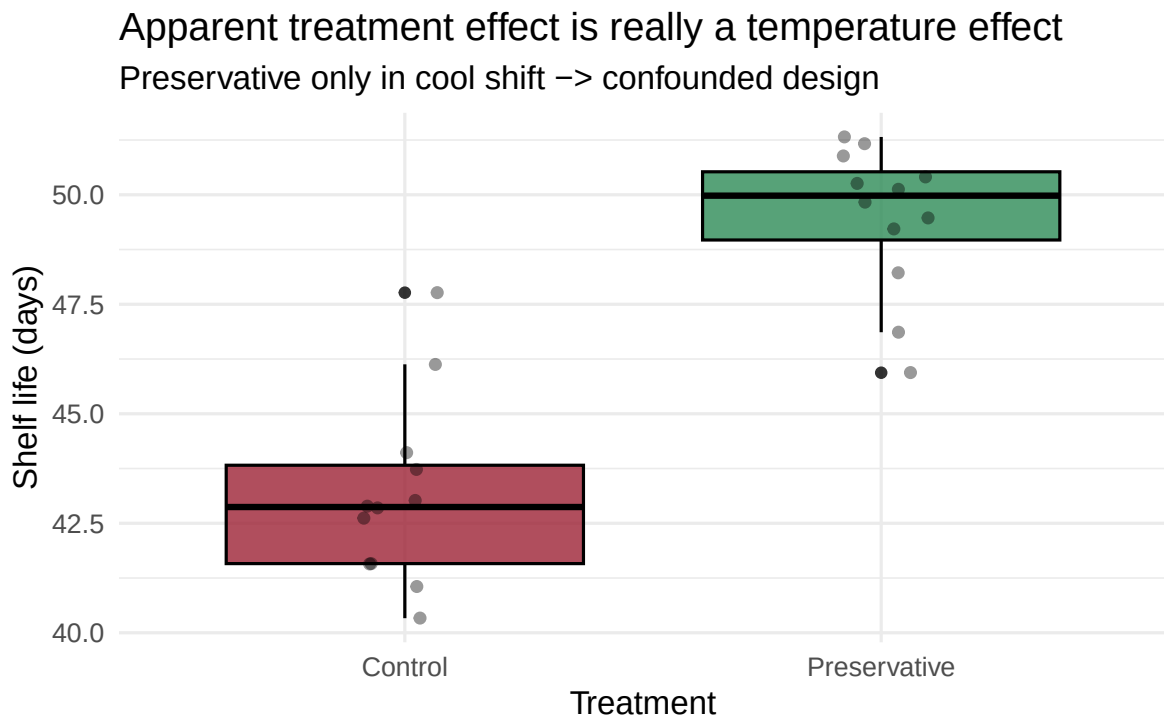


Figure 2: Confounding example: temperature trend mistaken for preservative effect

Timing: ~4 minutes

Explain: The “effect” vanishes if both treatments appear in both shifts. Reinforces randomization.

Replication: Biological vs Technical

- **Biological:** Different plants/animals/plots → captures real-world variability.
- **Technical:** Repeat measurement on the same sample → captures instrument noise.
- **Guideline:** Biological replication drives inference; technical replication improves precision but doesn’t replace biological reps.

Timing: ~3 minutes

Randomization: Practical Moves

- Shuffle treatment labels before field layout; don’t sort by convenience.
- Document the random seed and method for reproducibility.
- Possibly different levels of randomization.

The way you randomize your treatments defines the design (i.e., CRD, RCBD, etc.).

Timing: ~3 minutes

Tip: Demonstrate a quick randomization with a magic trick

Blocking: When and Why

- Use blocks when you can name a major nuisance source (slope, greenhouse bench, barn side).
- Blocks reduce residual noise → tighter confidence intervals.
- Don’t over-block: if blocks are too small, you lose power for treatments.

Timing: ~3 minutes

Agricultural Examples (Conceptual)

Field Example: Nitrogen on Wheat

- Treatments: 0, 60, 120 kg N/ha.
- Units: 12 plots on a mild slope.
- Design fix: Block by slope position (upper/mid/lower), randomize N within each block.
- Without blocking: Higher yields at lower slope falsely inflate N effect.

Timing: ~3 minutes

Livestock Example: Diet Trial

- Treatments: Control vs. protein-supplemented diet.
- Units: Pens of pigs; ventilation differs by barn side.
- Design fix: Block by barn side, randomize diets within side; replicate pens per diet.
- Without blocking: Barn-side effect masquerades as diet effect.

Timing: ~3 minutes

Food Quality Example: Pasteurization Study

- Treatments: Standard vs. slightly lower temperature.
- Units: Batches over two days; day-to-day equipment calibration drifts.
- Design fix: Randomize treatment order within each day; treat day as a block.
- Without blocking: Day effect confounds temperature effect.

Timing: ~3 minutes

Small-Group Activity (5 minutes)

Identify the Design Flaw

Scenario: A vineyard wants to test a canopy thinning practice. They thin only the sunny (south-facing) rows and compare to unthinned on the shaded (north-facing) rows.

- In pairs, list **two flaws** and propose a quick redesign.
- What would you randomize? What could you block?
- Share one fix with the class.

Timing: 5 minutes breakout + 2 minutes debrief

Likely answers: - Flaw: Treatment confounded with aspect/light. Redesign: Randomize thinning within paired adjacent rows; block by aspect or row pairs. - Flaw: Possible soil moisture gradient. Redesign: Blocks across moisture zones.

Looking Ahead

Summary & Key Takeaways

- Good design separates **treatment effects** from **nuisance variation**.
- Replication estimates noise; randomization protects against bias; blocking controls known gradients.
- Observational experimental; causation needs controlled assignment.
- Document your design decisions—future you (and reviewers) will thank you.

Timing: ~3 minutes

Next Lecture Preview

- Wednesday: Intro to AI as a study guide, R + Quarto basics; run your first R chunk.
- Bring: Laptop with VS Code, R/Quarto installed (or Posit Cloud account).
- We'll practice a simple randomization script together.

Timing: ~2 minutes

Motivation

“By the end of this semester, you’ll design better experiments than many published papers.”
— Because you’ll **plan**, **randomize**, **replicate**, and **block** with intent.

Timing: ~1 minute

Resources

- **Textbook:** Oehlert (2010), *A First Course in Design and Analysis of Experiments*, Chapter 1.
- **Data ideas:** agridat package (e.g., barley, wheat, livestock trials) for real agricultural examples.
- **Reading tip:** Focus on the why behind replication, randomization, blocking; equations come later.

Timing: ~2 minutes

Common questions & answers: - *“Do I always need blocks?”* Only when you can name a major nuisance source; otherwise randomize broadly and check later. - *“How many reps are enough?”* More is better until cost dominates; we’ll cover power later.